WEEK 6 REPORT

Healthcare Visualization & Dashboard

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Introduction

The objective of this week was to transform the healthcare data model developed in Week 5 into an interactive and informative dashboard using Power BI. Various visualization techniques were used to represent patient information, billing details, medical conditions, admission patterns, and insurance-related data.

The dashboard combines KPI cards, matrix tables, charts, and slicers to provide a clear overview of the healthcare dataset. Interactive filtering was also implemented so that users can explore the data based on insurance provider, admission type, and admission date.

KPI Summary Cards

Four KPI cards were created to display the important summary information of the healthcare dataset.

The following measures were used:

- Total Patients – Displays the total number of patients.
- Total Billing Amount – Shows the overall billing amount.
- Average Billing Amount – Represents the average billing amount per record.
- Average Length of Stay – Displays the average number of days patients stayed in the hospital.

These cards provide a quick overview of the major healthcare and billing indicators without requiring users to examine individual records.



Demographics and Patient Distribution

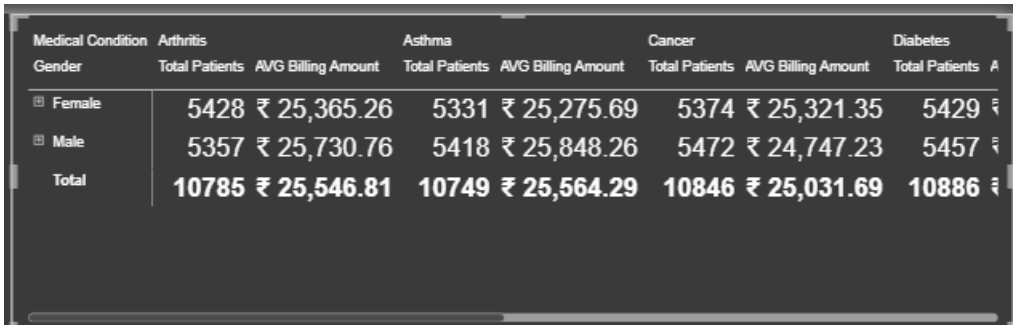
A Matrix visual was created to analyze patient distribution based on demographic and medical information.

The matrix uses:

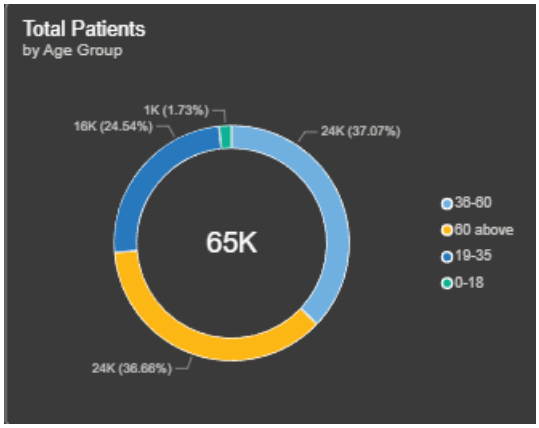
- Rows: Age Group and Gender
- Columns: Medical Condition
- Values: Total Patients and Average Billing Amount

This visualization allows patient counts and billing amounts to be compared across different age groups, genders, and medical conditions.

A Donut Chart was also created to represent the distribution of patients across different age groups. This provides an easy-to-understand visual comparison of patient populations.



Medical Condition	Arthritis		Asthma		Cancer		Diabetes	
Gender	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	A
Female	5428	₹ 25,365.26	5331	₹ 25,275.69	5374	₹ 25,321.35	5429	₹
Male	5357	₹ 25,730.76	5418	₹ 25,848.26	5472	₹ 24,747.23	5457	₹
Total	**10785**	**₹ 25,546.81**	**10749**	**₹ 25,564.29**	**10846**	**₹ 25,031.69**	**10886**	₹



Medical Conditions and Admission Volumes

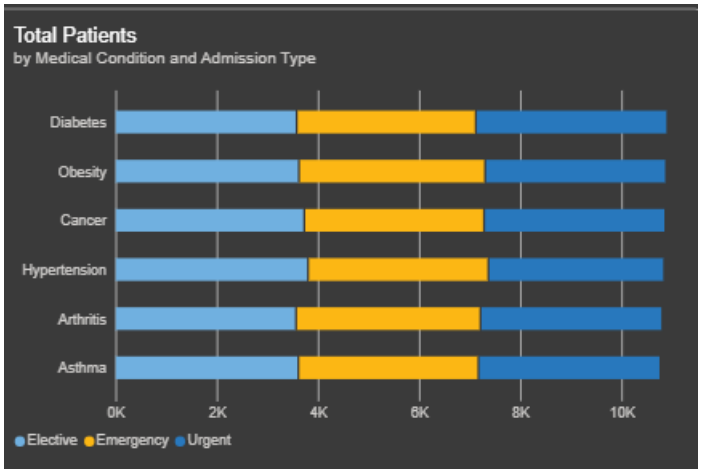
A Stacked Bar Chart was created to analyze the number of patients associated with different medical conditions.

The chart uses:

- Y-axis: Medical Condition
- X-axis: Total Patients
- Legend**:** Admission Type

The admission types are divided into Elective, Emergency, and Urgent categories. The stacked format makes it possible to compare the total number of patients for each medical condition while also observing how the patients are distributed among different admission types.

This visualization helps identify the relationship between medical conditions and the types of hospital admissions.



Admission Volume Trends and Hospital Analytics

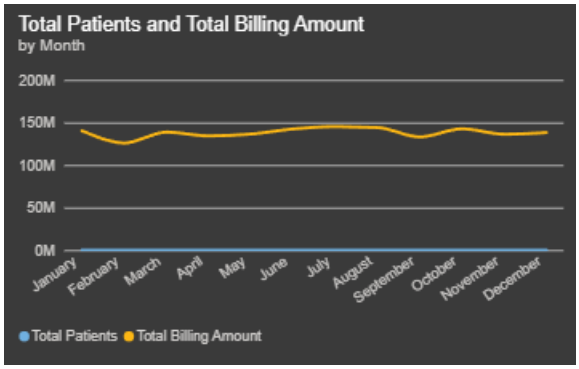
A Line Chart was created to visualize healthcare activity over time using the Date of Admission field.

The chart uses:

- X-axis: Date of Admission
- Values: Total Patients and Total Billing Amount

The date information was organized using the month level to observe changes in patient and billing activity over time.

This visualization helps provide a time-based view of healthcare activity and allows changes in patient volume and billing amounts to be observed.



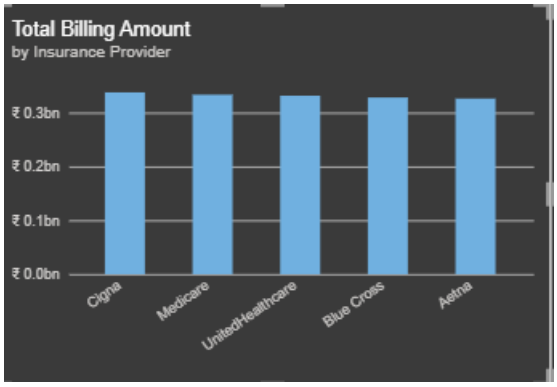
Insurance Provider Analysis

A Clustered Column Chart was created to compare the total billing amount associated with different insurance providers.

The chart uses:

- X-axis: Insurance Provider
- Y-axis: Total Billing Amount

The visualization makes it easier to compare the billing contribution of different insurance providers and identify differences in the overall billing amounts.



Slicers and Dynamic Filtering

Three slicers were added to make the dashboard interactive.

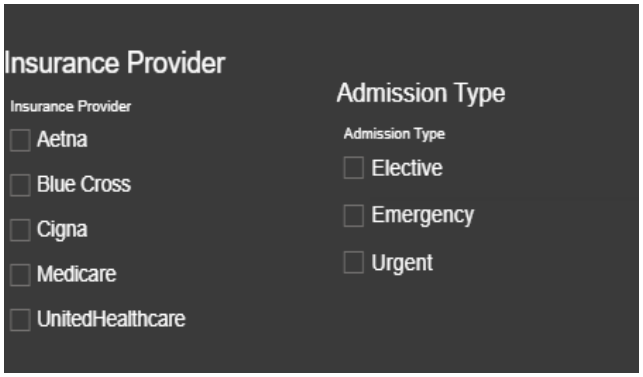
The slicers are:

1. Insurance Provider
2. Admission Type
3. Date of Admission

The Insurance Provider and Admission Type slicers allow users to select specific categories, while the Date of Admission slicer allows users to filter the dashboard based on a selected date range.

These slicers provide dynamic cross-filtering, meaning that selecting a value automatically updates the related charts and KPI cards. This makes the dashboard more flexible and allows users to analyze specific portions of the healthcare dataset.





Interactive Dashboard

All the visualizations were arranged together to create a single healthcare dashboard. The dashboard combines numerical summaries, demographic analysis, medical condition analysis, admission patterns, billing trends, and insurance provider information.

The interactive elements allow users to explore the data without manually changing individual charts. For example, selecting an insurance provider from the slicer updates the relevant visualizations based on that selection.

The final dashboard therefore provides both a **high-level summary** and the ability to perform more detailed analysis through filtering.



Key Outcomes

The main outcomes of the Week 6 implementation are:

- Created KPI cards for important healthcare measures.
- Developed a matrix for demographic and medical-condition analysis.
- Visualized patient age-group distribution using a donut chart.
- Compared medical conditions and admission types using a stacked bar chart.
- Analyzed admission and billing trends using a line chart.
- Compared insurance providers using a clustered column chart.
- Added interactive slicers for insurance provider, admission type, and admission date.
- Created a consolidated and interactive healthcare dashboard in Power BI.

Conclusion

The Week 6 task successfully converted the healthcare data model developed in the previous week into an interactive Power BI dashboard. Different visualization techniques were used to represent patient demographics, medical conditions, admission types, billing information, and insurance provider data in a simple and understandable manner.

The use of KPI cards provides quick access to important summary values, while charts and the matrix provide more detailed analysis. The addition of slicers further improves the dashboard by allowing users to dynamically filter and explore the data.

Overall, the completed dashboard demonstrates how Power BI can be used to transform structured healthcare data into meaningful visual insights and an interactive analytical report.